KAIZEN MATHS

Lesson starter

Limits and First Principles

Core Limits

Name _____

S1

Evaluate each limit by direct substitution.

$\lim_{x \to 0} 2x^2 + 3x - 8$

S2

Simplify each expression before taking the limit.

$\lim_{x \to -3} \dfrac{x^2 + 4x + 3}{x + 3}$

S3

Rationalise the numerator before taking the limit.

$\lim_{x \to 6} \dfrac{\sqrt{x + 3} - 3}{x - 6}$

S4

Use limit laws to evaluate each limit.

$\lim_{x \to 1} \dfrac{4x + 2}{3x + 2}$

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Show your working

KAIZEN MATHS

Exit ticket

Limits and First Principles

Core Limits

Name _____

E1

Evaluate each limit by direct substitution.

$\lim_{x \to 2} -2x^2 - x - 1$

E2

Simplify each expression before taking the limit.

$\lim_{x \to 3} \dfrac{x^2 - 4x + 3}{x - 3}$

E3

Rationalise the numerator before taking the limit.

$\lim_{x \to 5} \dfrac{\sqrt{x + 4} - 3}{x - 5}$

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Simplify each expression before taking the limit.

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Show your working

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Limits and First Principles

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Evaluate each limit by direct substitution.

$\lim_{x \to 2} -2x^2 - x - 1$

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Simplify each expression before taking the limit.

$\lim_{x \to 3} \dfrac{x^2 - 4x + 3}{x - 3}$

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Rationalise the numerator before taking the limit.

$\lim_{x \to 5} \dfrac{\sqrt{x + 4} - 3}{x - 5}$

ALG-911D-L1 · E

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Core Limits

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Evaluate each limit by direct substitution.

$\lim_{x \to 2} -2x^2 - x - 1$

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Simplify each expression before taking the limit.

$\lim_{x \to 3} \dfrac{x^2 - 4x + 3}{x - 3}$

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Rationalise the numerator before taking the limit.

$\lim_{x \to 5} \dfrac{\sqrt{x + 4} - 3}{x - 5}$

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Teacher answers & working

Limits and First Principles · Core Limits

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S1

Evaluate each limit by direct substitution.

$\lim_{x \to 0} 2x^2 + 3x - 8$

Answer:

-8

Working:

Polynomials are continuous, so this limit can be evaluated by substitution.

$$\begin{aligned}\lim_{x \to 0} 2x^2 + 3x - 8 &= 2(0)^2 + 3(0) - 8 \\ &= 2(0)^2 - 8 \\ &= -8\end{aligned}$$

The value approached by the function is -8.

S2

Simplify each expression before taking the limit.

$\lim_{x \to -3} \dfrac{x^2 + 4x + 3}{x + 3}$

Answer:

-2

Working:

Direct substitution gives a zero denominator, so factorise before evaluating.

$$\begin{aligned}\lim_{x \to -3} \frac{x^2 + 4x + 3}{x + 3} &= \lim_{x \to -3} \frac{(x + 3)(x + 1)}{x + 3} \\ &= \lim_{x \to -3} (x + 1)\end{aligned}$$

After cancelling, the simplified expression is valid for values close to $x = -3$, with $x \neq -3$.

$$\begin{aligned}\lim_{x \to -3} (x + 1) &= -3 + 1 \\ &= -2\end{aligned}$$

Teacher answers & working

Limits and First Principles · Core Limits

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S3

Rationalise the numerator before taking the limit.

$$\lim_{x\to 6}\frac{\sqrt{x+3}-3}{x-6}$$

Answer:

$$\frac{1}{6}$$

Working:

Substitution gives $\frac{0}{0}$, so multiply by the conjugate.

$$\lim_{x\to 6}\frac{\sqrt{x+3}-3}{x-6}=\lim_{x\to 6}\frac{\sqrt{x+3}-3}{x-6}\cdot\frac{\sqrt{x+3}+3}{\sqrt{x+3}+3}$$

$$\lim_{x\to 6}\frac{x+3-9}{(x-6)(\sqrt{x+3}+3)}$$

$$=\lim_{x\to 6}\frac{x-6}{(x-6)(\sqrt{x+3}+3)}$$

$$\lim_{x\to 6}\frac{1}{\sqrt{x+3}+3}$$

$$\lim=\frac{1}{\sqrt{9}+3}$$

$$=\frac{1}{3+3}$$

$$=\frac{1}{6}$$

Teacher answers & working

Limits and First Principles · Core Limits

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S4

Use limit laws to evaluate each limit.

$$\lim_{x \to 1} \frac{4x + 2}{3x + 2}$$

Answer:

$$\frac{6}{5}$$

Working:

$$\lim_{x \to 1} \frac{4x + 2}{3x + 2} = \frac{\lim_{x \to 1}(4x + 2)}{\lim_{x \to 1}(3x + 2)}$$

The denominator does not approach zero, so the quotient law can be used.

$$\begin{aligned} &= \frac{4(1) + 2}{3(1) + 2} \\ &= \frac{6}{5} \\ &= \frac{6}{5} \end{aligned}$$

E1

Evaluate each limit by direct substitution.

$$\lim_{x \to 2} -2x^2 - x - 1$$

Answer:

$$-11$$

Working:

$$\lim_{x \to 2} -2x^2 - x - 1 = -2(2)^2 - (2) - 1$$

Polynomials are continuous, so this limit can be evaluated by substitution.

$$\begin{aligned} &= -2(2)^2 - 2 - 1 \\ &= -11 \end{aligned}$$

The value approached by the function is -11.

Teacher answers & working

Limits and First Principles · Core Limits

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E2

Simplify each expression before taking the limit.

$$\lim_{x\to 3}\frac{x^2-4x+3}{x-3}$$

Answer:

2

Working:

Direct substitution gives a zero denominator, so factorise before evaluating.

$$\lim_{x\to 3}\frac{x^2-4x+3}{x-3}=\lim_{x\to 3}\frac{(x-3)(x-1)}{x-3}$$
$$=\lim_{x\to 3}(x-1)$$

After cancelling, the simplified expression is valid for values close to $x=3$, with $x\neq 3$.

$$\lim_{x\to 3}(x-1)=3-1$$
$$=2$$

E3

Rationalise the numerator before taking the limit.

$$\lim_{x\to 5}\frac{\sqrt{x+4}-3}{x-5}$$

Answer:

$\frac{1}{6}$

Working:

Substitution gives $\frac{0}{0}$, so multiply by the conjugate.

$$\lim_{x\to 5}\frac{\sqrt{x+4}-3}{x-5}=\lim_{x\to 5}\frac{\sqrt{x+4}-3}{x-5}\cdot\frac{\sqrt{x+4}+3}{\sqrt{x+4}+3}$$
$$\lim_{x\to 5}\frac{x+4-9}{(x-5)(\sqrt{x+4}+3)}$$
$$=\lim_{x\to 5}\frac{x-5}{(x-5)(\sqrt{x+4}+3)}$$
$$\lim_{x\to 5}\frac{1}{\sqrt{x+4}+3}$$
$$\lim=\frac{1}{\sqrt{9}+3}$$
$$=\frac{1}{3+3}$$
$$=\frac{1}{6}$$